

Mathis DELSART - AI/ML Engineer

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SUMMARY

Machine Learning & AI Engineer designing, building, and deploying production-minded AI systems end-to-end. Experience spans distributed training on HPC GPU clusters (SLURM), Docker-containerized cloud deployment (Azure), and full-stack delivery from frontend to scalable, API-driven backends. Passionate about Computer Vision, Natural Language Processing, Generative AI (LLMs & RAG), and Agentic AI.

EDUCATION

MSc IN COMPUTER SCIENCE AND ENGINEERING

École Polytechnique de Louvain, UCLouvain

Major in Artificial Intelligence

Graduated with 17.95/20, **SUMMA CUM LAUDE**

Thesis: "Deep Reinforcement Learning for Competitive Agents in MicroRTS: Architecture, Training, and Tournament Evaluation"

Sept. 2024 - June 2026

Louvain-la-Neuve, Belgium

BSc IN ENGINEERING

École Polytechnique de Louvain, UCLouvain

Major in Computer Science & Electrical Engineering

Graduated with 16.74/20, **MAGNA CUM LAUDE**

Sept. 2021 - June 2024

Louvain-la-Neuve, Belgium

PUBLICATIONS

M. DELSART, A. MORENVILLE, É. PIETTE, "Combining Spatial and Entity-Based Reasoning for Competitive MicroRTS via U-Net and Transformers", *IEEE Conference on Games (CoG)*, Sept. 2026, Madrid, Spain — **accepted**

Sept. 2026

Madrid, Spain

PROFESSIONAL EXPERIENCE

PRIVATE TUTOR

Mathematics, Physics & Programming

Taught 100+ students over 3+ years in math, physics & programming, sharpening my ability to explain technical concepts clearly.

Built and deployed a website (mathisdelsart.github.io/private-tutoring) to grow and manage the activity.

2022 - Present

Hainaut, Belgium

PROJECT EXPERIENCE

GROUNDED RAG — AGENTIC STUDY ASSISTANT

Agentic RAG with per-source citations and a faithfulness guard

Built a RAG assistant over course material (math-aware PDF extraction) that cites every answer by chapter/page and refuses out-of-source questions via a faithfulness check.

A LangGraph router generates and auto-grades exercises (LLM-as-a-judge), served via

FastAPI with LangFuse tracing and a CI eval set.

June 2026

Python | LangGraph | FastAPI | Qdrant | Docker | LangFuse

DRL AGENT FOR COMPETITIVE MICRORTS

Custom policy: U-Net + entity-level Transformer (PPO)

UECD policy (U-Net · Entity · CBAM · Deep) trained with PPO via self-play and opponent curricula.

Extended the Java↔Python env (20+ feature mechanisms) and built a reproducible tournament framework.

Topped a 19-agent tournament at 96.67% WR and 65.7% H2H vs prior DRL winner at $\sim 2.5\times$ less compute.

Results, replays & analyses: mathisdelsart.github.io/microrts-drl-uecd-website

Sept. 2025 - June 2026

Python | PyTorch | SLURM | Java | JPytype

SCALABLE CLOUD-NATIVE E-COMMERCE PLATFORM

Distributed microservices with elastic auto-scaling

Built a microservices platform with 8 backend services behind an API gateway, featuring

JWT authentication and a MapReduce-based recommendation engine. Deployed on Azure VMs

with Docker Swarm auto-scaling, Redis job queues, and Azure Blob Storage for product images.

Oct. - Dec. 2025

Node.js | SvelteKit | Docker Swarm | Azure | CouchDB

NLP PIPELINE: CLASSIFICATION, EMBEDDINGS & QUESTION ANSWERING

From probabilistic models (Naive Bayes) to transformers (BERT)

Built Naive Bayes classification reaching $\sim 80\%$ accuracy on 60k Stack Overflow questions,

benchmarked sparse vs dense embeddings (TF-IDF, PPMI, Word2Vec), and fine-tuned BERT on

SQuAD (Stanford Question Answering Dataset) for extractive reading comprehension.

Oct. - Dec. 2024

Python | PyTorch | Hugging Face | Gensim | NLTK

SKILLS

TECHNICAL SKILLS

ML / DL Python, PyTorch, scikit-learn, Hugging Face, NumPy, Pandas
WEB React, Next.js, Node.js, FastAPI, Tailwind CSS, TypeScript
DEVOPS Git, CI/CD, Docker, Docker Swarm, Azure, SLURM, Bash

GENAI LangChain, LangGraph, RAG
DATABASES PostgreSQL, SQLite, CouchDB, Qdrant, Redis
AI TOOLING GitHub Copilot, Claude Code

SOFT SKILLS

INTERPERSONAL Communication, Mentoring, Technical Vulgarization
MINDSET Analytical Rigor, Problem-solving, Continuous Learning
ORGANIZATION Time Management, Prioritization, Adaptability

COLLABORATION Teamwork, Code Review, Knowledge Sharing
WORK ETHIC Autonomy, Initiative, Perseverance

LANGUAGES

English Professional Working Proficiency | French Native